will form—unless there is yet another stellar graveyard, unsuspected in the 1930s.

This line of reasoning is so obvious in retrospect that it seems amazing that Zwicky did not pursue it, Chandrasekhar did not pursue it, Eddington did not pursue it. Had Zwicky tried to pursue it, however, he would not have got far; he understood too little nuclear physics and too little relativity to be able to discover whether the laws of physics place a mass limit on neutron stars or not. At Caltech there were, however, two others who *did* understand the physics well enough to deduce neutron-star masses: Richard Chace Tolman, a chemist turned physicist who had written a classic textbook called *Relativity, Thermodynamics, and Cosmology;* and J. Robert Oppenheimer, who would later lead the American effort to develop the atomic bomb.

Tolman and Oppenheimer, however, paid no attention at all to Zwicky's neutron stars. They paid no attention, that is, until 1938, when the idea of a neutron star was published (under the slightly different name of *neutron core*) by somebody else, somebody whom, unlike Zwicky, they respected: Lev Davidovich Landau, in Moscow.

Landau

Landau's publication on neutron cores was actually a cry for help: Stalin's purges were in full swing in the U.S.S.R., and Landau was in danger. Landau hoped that by making a big splash in the newspapers with his neutron-core idea, he might protect himself from arrest and death. But of this, Tolman and Oppenheimer knew nothing.

Landau was in danger because of his past contacts with Western scientists:

Soon after the Russian revolution, science had been targeted for special attention by the new Communist leadership. Lenin himself had pushed a resolution through the Eighth Congress of the Bolshevik party in 1919 exempting scientists from requirements for ideological purity: "The Problem of industrial and economic development demands the immediate and widespread use of experts in science and technology whom we have inherited from capitalism, despite the fact that they inevitably are contaminated with bourgeois ideas and customs." Of special concern to the